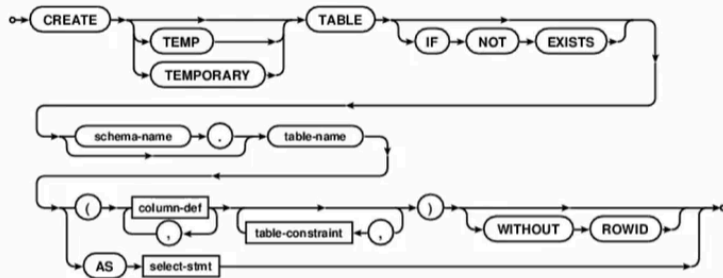


Lecture 34.Database

Create Table and Drop Table

Create Table

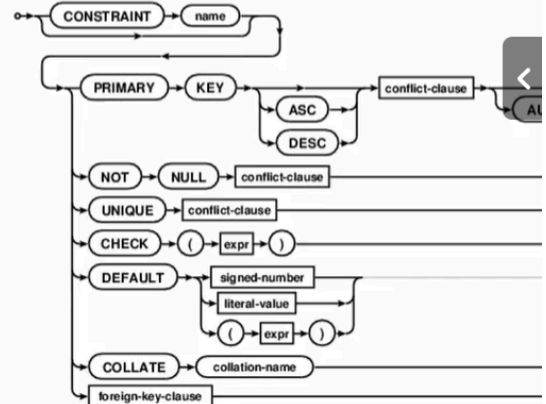
CREATE TABLE expression syntax:



column-def:

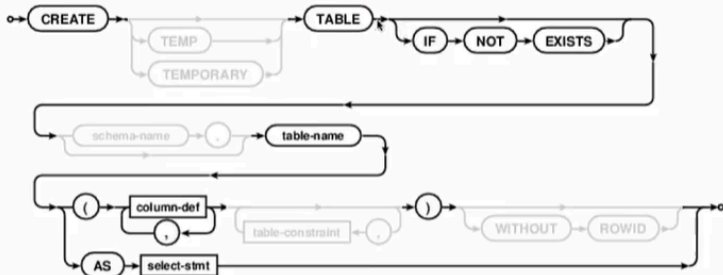


column-constraint:



Create Table

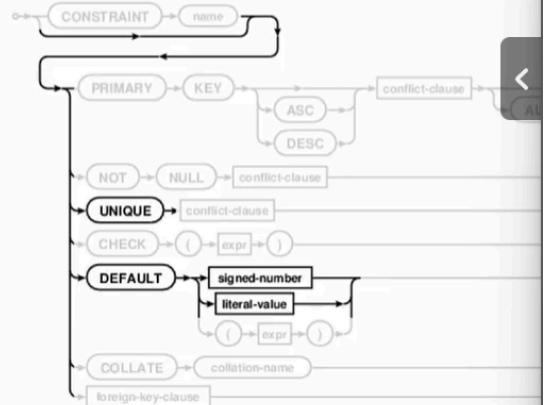
CREATE TABLE expression syntax:



column-def:

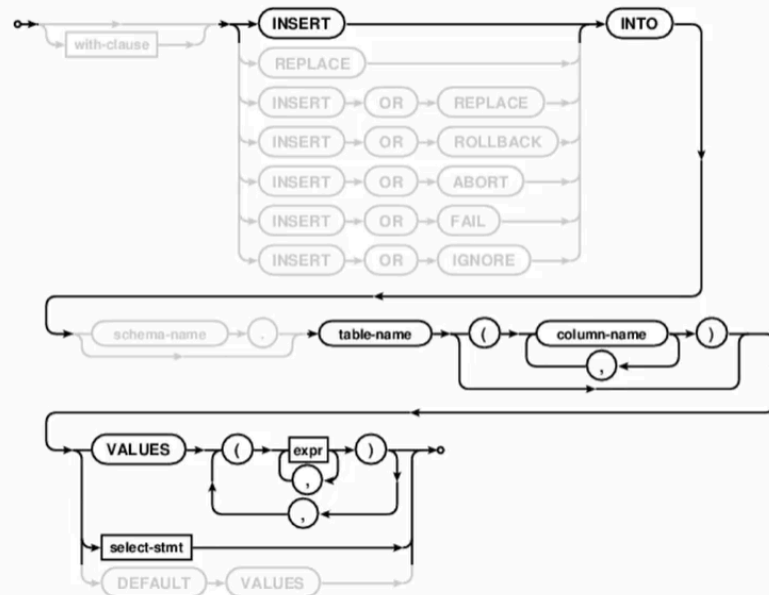


column-constraint:



Modifying Tables

Insert



For a table t with two columns...

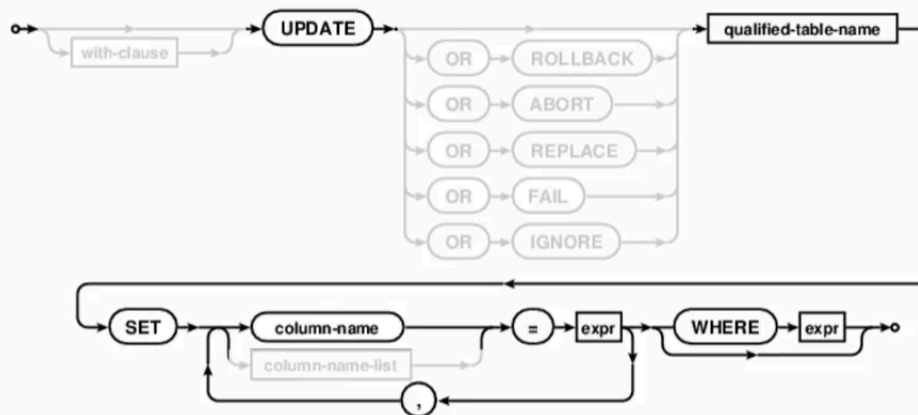
To insert into one column:

```
INSERT INTO t(column) VALUES (value);
```

To insert into both columns:

```
INSERT INTO t VALUES (value0, value1);
```

Update



Update sets all entries in certain columns to new values, just for some subset of rows.

Python and SQL

```
import sqlite3

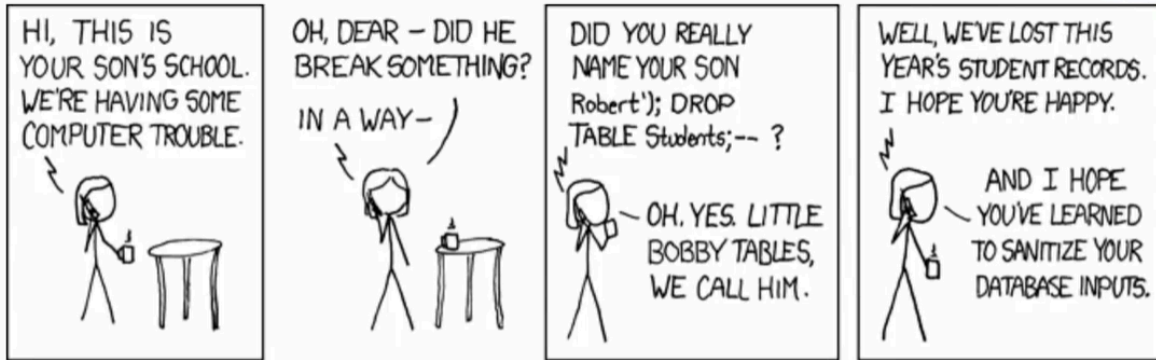
# connect to a sql file
sqlite3.Connection("n.db")

# execute inside the sql file
db.execute("CREATE TABLE nums SELECT 2 UNION SELECT 3;")
db.execute("INSERT INTO nums VALUE (?), (?), (?), range(4, 7))")
# fetch the content and turn it into tuple in python
tuple = db.execute("SELECT * FROM nums;").fetchall()

# commit the changes
db.commit()
```

SQL Injection Attack

A Program Vulnerable to a SQL Injection Attack



```
name = "Robert'); DROP TABLE Students; --"
```

```
cmd = "INSERT INTO Students VALUES ('" + name + "')";
```

```
db.executeScript(cmd) db.execute("INSERT INTO Students VALUES (?)", [name])
```

```
INSERT INTO Students VALUES ('Robert'); DROP TABLE Students; --');
```

```
INSERT INTO Students VALUES ('Robert'); DROP TABLE Students; --');
```

<https://xkcd.com/327/>